Name ______________________________________ **Date** ______________

Worksheet 2.1.1

Materials: properties and changes

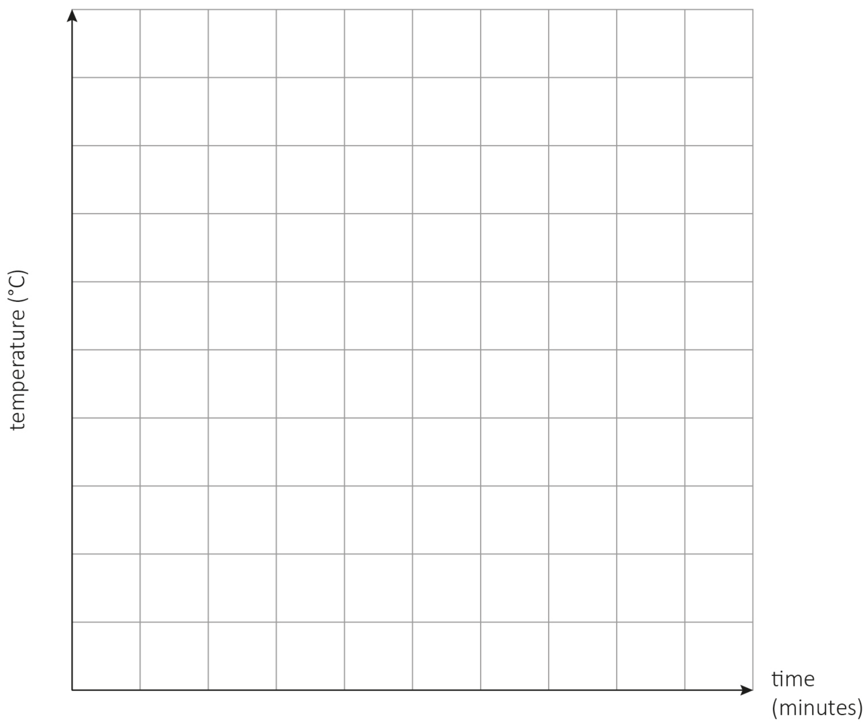
Class 6 melted ice in a saucer and measured its temperature. These are their results.

Time (minutes)	0	2	4	6	8	10	12	14
Temperature (°C)	-6	-4	-2	0	0	0	2	4

1 a What kind of graph would you draw to show these results?

____

b Draw the graph.



2 a Which temperature is the melting point of ice?

____

b Do any other substances melt at this temperature? Say why or why not.

____

____

c Why does the ice on the saucer melt? Use the particle model to explain your answer.

____

____

____

____

Help sheet

We use bar graphs to compare things between different groups or objects, for example, the melting points of different solids.

We draw line graphs to track changes over short and long periods of time. For example, measuring the change in temperature of water over time as we heat the water. Line graphs can also be used to compare changes over the same period of time for more than one group. For example, comparing the time it takes for different quantities of water to heat up to boiling point.

Name ____ **Date** _______________

Stretch questions

3 a For how long does the does the temperature stay the same?

____

b Suggest a reason for this.

____

____

4 a Predict the water temperature after 20 minutes.

____

b Show how you obtained your prediction by drawing onto the graph.

Name ________________________________ **Date** ______________

Worksheet 2.1.2A

Materials: properties and changes

Mrs Kaur investigated boiling with her class. She heated water and measured its temperature.
The class recorded these results.

Time (minutes)	2	4	6	8	10	12	14
Temperature (°C)	30	45	60	75	90	100	100

1 Draw a line graph of the results.

2 a What is the boiling point of water?

____

b How could you find out if water always boils at this temperature?

____

____

Name ______________________________________ **Date** ______________

Worksheet 2.1.2B

Materials: properties and changes

Mrs Kaur investigated boiling with her class. She heated water and measured its temperature. The class recorded these results.

Time (minutes)	2	4	6	8	10	12	14
Temperature (°C)	30	45	60	75	90	100	100

Mr Moola's class carried out the same investigation. These are their results.

Time (minutes)	2	4	6	8	10	12	14
Temperature (°C)	40	60	80	100	100	100	100

1 Draw a line graph for each set of results.

2 a What is the boiling point of water?

____

b Do the two graphs support this conclusion? Explain your answer.

____

____

c How could you find out if water always boils at this temperature?

____

____

3 Suggest two ways you would know when the water boiled if you did not measure the temperature.

____

____

Name ______________________________________ **Date** ______________

Worksheet 2.1.2C

Materials: properties and changes

Mrs Kaur investigated boiling with her class. She heated water and measured its temperature.
The class recorded these results.

Time (minutes)	2	4	6	8	10	12	14
Temperature (°C)	30	45	60	75	90	100	100

Mr Moola's class carried out the same investigation. These are their results.

Time (minutes)	2	4	6	8	10	12	14
Temperature (°C)	40	60	80	100	100	100	100

1 Draw a line graph for each set of results.

2 a What is the boiling point of water?

____

b Do the two graphs support this conclusion? Explain your answer.

____

____

c How could you find out if water always boils at this temperature?

____

____

3 Suggest two ways you would know when the water boiled if you did not measure the temperature.

____

____

4 a Are the graphs identical?

____

b What difference do you notice in the patterns of the two graphs?

____

____

____

c Suggest a reason for this difference.

____

____

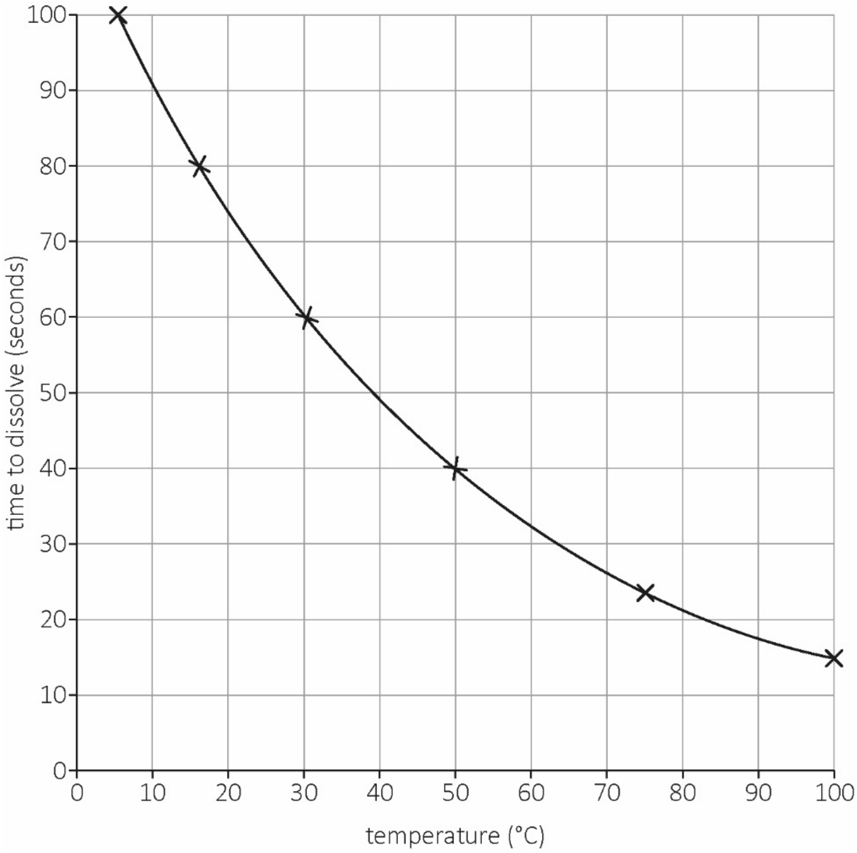
____

Name ______________________________________ **Date** ______________

Worksheet 2.3

Reversible changes

Class 6 conducted a fair test to find out how different temperatures affect the rate at which sugar dissolves in water. They drew a graph of their results.



1 a Name the solvent used.

____

b Name the solute used.

____

2 a At which temperature did the sugar dissolve fastest?

____

b How long did it take the sugar to dissolve at 30 °C?

____

c Predict how long it would take the sugar to dissolve at 20 °C.

____

d At which temperature did the sugar dissolve slowest?

____

3 In order to make the investigation a fair test, name:

a the control variables that Class 6 kept the same

____

____

b the dependent variable that they measured

____

c the independent variable that they changed.

____

4 Write a conclusion for the investigation.

____

____

Help sheet

Remember that a solution is made of two parts:

- the solute – the substance that dissolves, usually a solid, e.g. salt or sugar
- the solvent – the substance in which the solute dissolves, usually a liquid, e.g. water.

When we read a line graph, we look at two sets of data that are connected, such as water temperature and how fast sugar dissolves. The graph shows us how one thing or factor changes in relation the other. The labels on the axes tell you which factors you are looking at.

We can find information about both factors from the graph. We need to look at the data points plotted. To find the time taken for sugar to dissolve at a certain temperature, find the temperature on the x-axis, then look for the data point on the graph line for that temperature. Now look at the value on the y-axis for that data point. This will be the time taken for the sugar to dissolve at that temperature.

Name ______________________________ **Date** ______________

Stretch questions

5 Class 6 repeated their measurements of the time taken for sugar to dissolve at different temperatures. Their results are recorded in the table.

Temperature in °C	Time for sugar to dissolve in seconds	Time for sugar to dissolve in seconds	Time for sugar to dissolve in seconds	Average time for sugar to dissolve in seconds
10	77	73	75	
20	52	48	50	
30	34	36	32	
40	26	27	19	
50	19	18	17	

a Why did class 6 repeat their measurements?

____

____

b What overall pattern can you see in the results?

____

____

c Are there any results that do not fit the pattern? If so, which results?

____

____

d Calculate the average for the different temperatures. Fill in the last column of the table.

e Draw a line graph of the results.

f At which temperature did the sugar dissolve

fastest? ________________

slowest? ________________

g Explain your answers to question 5f.

____

____

____

____

h Predict the time it would take for the sugar to dissolve at a temperature of 80 °C.

Plot this data point on your graph.

Name ______________________________________ **Date** ______________

Worksheet 2.4

Chemical reactions

Plaster of Paris is a white powder that you mix with water. Doctors use it to help broken arms and legs grow back together properly. You can also make models with it.

Class 6 used plaster of Paris to investigate evidence for chemical reactions.

They mixed 75 ml of plaster of Paris powder with different quantities of water at a temperature of 35 °C.

They measured the temperature every 15 minutes for one hour.

The table shows their results.

Group	Amount of water used	Temperature in °C				
		Start	15 min	30 min	45 min	60 min
1	75 ml	35	37	40	44	39
2	100 ml	35	38	44	50	40
3	150 ml	35	40	50	60	45
4	200 ml	35	37	41	45	41
5	250 ml	35	36	38	40	37

1 a Identify the variables in the investigation.

Dependent variable ________________

Independent variable ________________

Control variables ________________

b Was the test fair or not? Say why or why not.

____

____

____

2 a What evidence for a chemical reaction does the investigation show?

____

b Name two other types of evidence for a chemical reaction.

____

____

3 How does the quantity of water added to the plaster of Paris affect the highest temperature reached? Describe the pattern you see in the results.

____

____

____

4 When we dissolve Epsom salts in water the temperature decreases.

Is the change in temperature evidence of a chemical reaction? Say why or why not.

____

____

____

Help sheet

To tell if a chemical reaction has taken place, you can ask these questions:

- Can you see gas bubbles?
- Can you see a change in colour?
- Does the temperature increase or decrease?
- Does a new substance form?

Name ______________________________________ **Date** ______________

Stretch questions

5 Why do you think the temperature did not change much for the first 15 minutes? (Think about the particle model.)

____

____

____

6 Why do you think the temperatures recorded increase and then decrease?

____

____

____

____

____